# Linked List

Fundamentals of Programming – CSC10012

Hồ Tuấn Thanh

htthanh@fit.hcmus.edu.vn

# Contents

- Linked List Concepts.
- Linked List Operations.
- Linked List Improvement.

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- **Linked List Concepts.**
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# Linked List Concepts

- Limitations of Array:
  - Characteristics:

    Continuous memory storage.

| | $a_0$ | $a_1$ | $a_2$ | $a_3$ | $a_4$ | $a_5$ | $a_6$ | $a_7$ | $a_8$ | $a_9$ |
|---|---|---|---|---|---|---|---|---|---|---|
| **a** | ? | ? | ? | ? | ? | ? | ? | ? | ? | ? |

  - Advantages:

    Random element access by indexing.

    Very efficient for fix-sized storage.
  - Disadvantages:

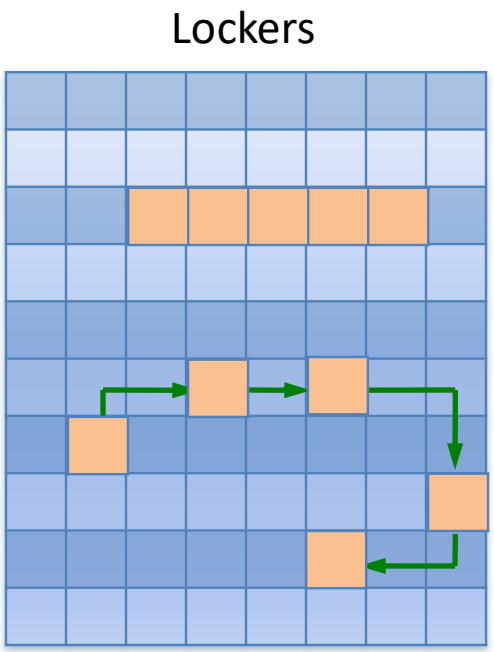
    Resize array requires memory reallocation.

    Add or remove element need element shifted.

    Allocate large continuous memory is difficult.

# Linked List Concepts

- Linked list solution:
  - Locker renting problem:
    - Need to rent lockers to store N items.
    - Each locker keeps only 1 item.
  - Continuous solution→ Array.
  - Dis-continuous solution:
    - Rent N arbitrary lockers.
    - Each locker keeps:
      - 1 items.
      - Address of next locker.
    - Only keep address of first locker.



# Linked List Concepts

- Singly linked list:
  - A discontinuous data structure.
  - Element = Data + Link to next element.
  - Last element link to NULL.
  - Head: link to first element.

  - C declaration:

```
struct Node
{
  int        data;
  Node       *next;
};
```

```
struct List
{
        Node     *head;
};
```

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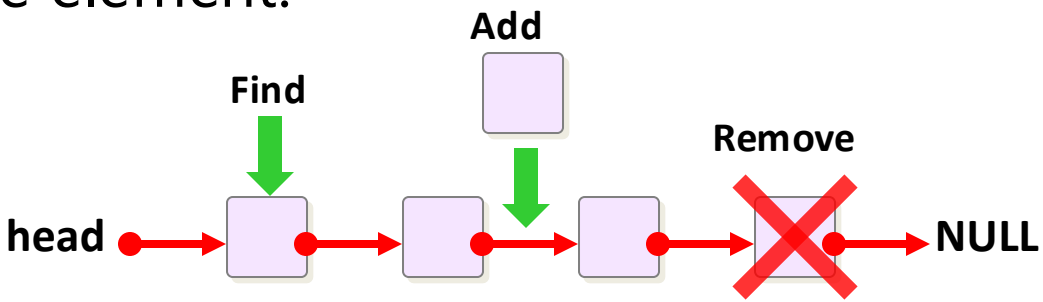
# Linked List Operations

- Operations on linked list:
  - Initialize.
  - Check empty.
  - Find element.
  - Add element.
  - Remove element.



# Linked List Operations

- Initialize list:
  - At first, list has no element.

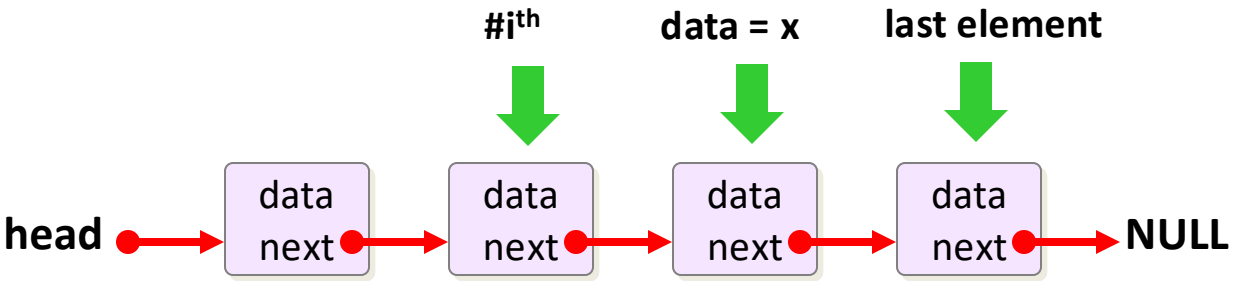
    **head** → **NULL**

- Check empty list:
  - An empty list has no element.

    **head** → **NULL ???**

# Linked List Operations

- Find element:
  - Find #i$^{th}$ element.
  - Find element has data x.
  - Find last element.

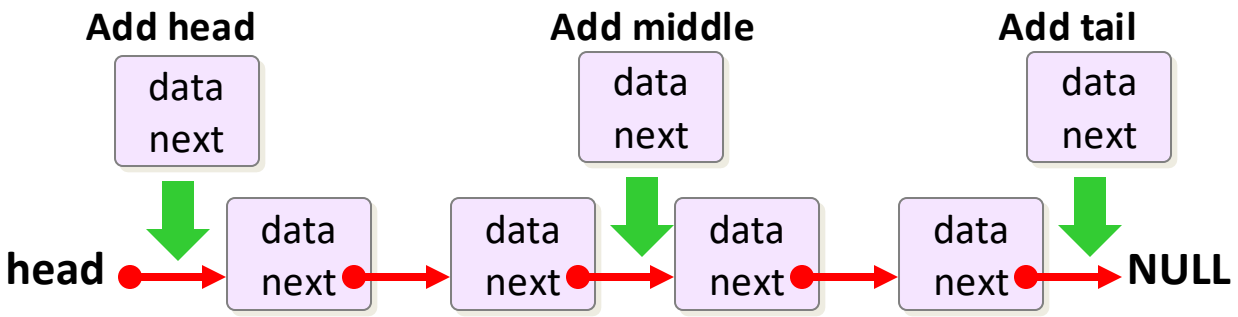


# Linked List Operations

- Add element:
  - Add head.
  - Add tail.
  - Add middle:

    After #i$^{th}$ element.

    Keep order (ascending).

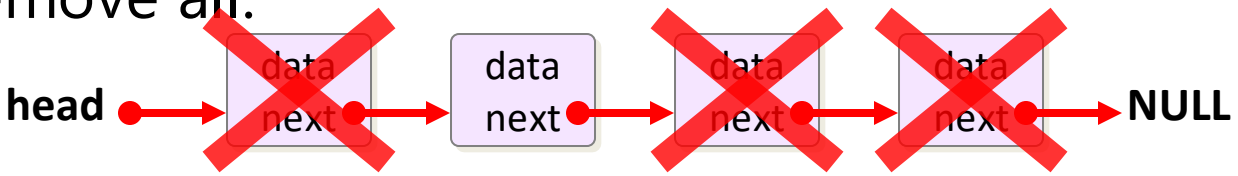


# Linked List Operations

- Remove element:
  - Remove head.
  - Remove tail.
  - Remove middle:

    #i$^{th}$ element.

    Element has data x.
  - Remove all.



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- Linked List Operations.
- **Linked List Improvement.**

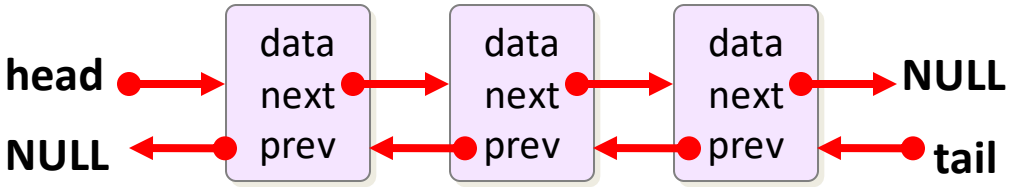
# Linked List Improvement

| | Dynamic array | Singly linked List |
|---|---|---|
| Organization | Continuous | Dis-continuous |
| Resize | Require re-allocation<br>Complexity: **O( n )** | No re-allocation<br>Complexity: **O( 1 )** |
| Access element | Random access<br>Complexity: **O( 1 )** | Sequential access<br>Complexity: **O( n )** |
| Search | Forward/Backward<br>Complexity: **O( n )** | Forward only<br>Complexity: **O ( n )** |
| Add/Remove element | Require element shifting<br>Complexity: **O( n )** | No shifting<br>Complexity: **O( 1 )** |
| Memory cost | No extra memory | Require extra memory<br>Cost: **8 * n bytes** |

➔ Efficient way to store ***sequential*** and ***variable-size*** data.

# Linked List Improvement

- Doubly linked list:
  - Element = Data + Link to next + Link to previous.
  - Head: forward traverse.
  - Tail: backward traverse.




  - C declaration:

```
struct DNode
{
  int       data;
  DNode     *next;
  DNode     *prev;
};
```

```
struct DList
{
        DNode   *head;
        DNode   *tail;
};
```

# Linked List Improvement

| | Dynamic array | Doubly linked list |
|---|---|---|
| Organization | Continuous | Dis-continuous |
| Resize | Require re-allocation<br>Complexity: **O( n )** | No re-allocation<br>Complexity: **O( 1 )** |
| Access element | Random access<br>Complexity: **O( 1 )** | Sequential access<br>Complexity: **O( n )** |
| Search | Forward/Backward<br>Complexity: **O( n )** | Forward/Backward<br>Complexity: **O ( n )** |
| Add/Remove element | Require element shifting<br>Complexity: **O( n )** | No shifting<br>Complexity: **O( 1 )** |
| Memory cost | No extra memory | Require extra memory<br>Cost: **16 * n bytes** |

➔ Good for ***two-directional traverse***, ***variable-size*** data.

# Linked List Improvement

- Skip list:

  Set of singly linked lists organized in layers.

  Lower layer: fine-grained nodes, slower “lane”.

  Higher layer: coarse-grained nodes, faster “lane”.

# Linked List Improvement

| | Ordered Array | Skip List |
|---|---|---|
| Organization | Continuous | Dis-continuous |
| Resize | Require re-allocation<br>Complexity: **O( n )** | No re-allocation<br>Complexity: **O( 1 )** |
| Access element | Random access<br>Complexity: **O( 1 )** | Sequential access<br>Complexity: **O( log(n) )** |
| Binary search | Forward/Backward<br>Complexity: **O( n*log(n) )** | Forward only<br>Complexity: **O ( n*log(n) )** |
| Add/Remove element | Require element shifting<br>Complexity: **O( n )** | No shifting<br>Complexity: **O( 1 )** |
| Memory cost | No extra memory | Require extra memory<br>Cost: **log( n ) * 4 * n bytes** |

➔ Efficient way to store ***variable-size*** and ***ordered*** data.

# Summary

- Singly linked list concepts:
  - Dis-contiguous storage.
  - Node = data + next.
  - Last node points to NULL.
- Singly linked list operations:
  - Initialize, check empty.
  - Find, add, remove.
- Singly linked list improvements:
  - Doubly linked list.
  - Skip list.

# Practice

- Practice 10.1:

  Write C/C++ program to implement operations on Doubly linked list.

# Practice

- Practice 10.2:

  Write C/C++ program to do the following on Singly linked list:

  - Add new node (keep order).
  - Reverse list.
  - Remove duplicated nodes (keep one node).
  - Remove $k^{th}$ node from tail.
  - Check if list is symmetric.

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